

Failover Streaming Replication

How to Perform Manual Failover in PostgreSQL Streaming Replication (No repmgr)

Table of Contents

- 0. Objective
- 1. Environment Setup
- 2. Simulate and Confirm Primary (rac1) Failure
- 3. Failover Procedure: Promote rac2 (Standby1) as New Primary
 - 3.1 Promote rac2 to Primary
 - 3.2 Update pg_hba.conf
 - 3.3 Reset primary_conninfo in postgresql.auto.conf
 - 3.4 Restart PostgreSQL Service
- 4. Verify Failover Operation (rac2 is primary)
- 5. Reconfigure rac3 (Standby2) to Point to New Primary (rac2)
 - 5.1 Update primary_conninfo on rac3
 - 5.2 Update pg_hba.conf
 - 5.3 Restart PostgreSQL Service
- 6. Data Validation Test – Change 1
 - 6.1 Insert Data on New Primary and Verify logs
 - 6.2 Create replication slot (rac3_standby_slot') on New Primary
 - 6.3 Verify Data on Standby and logs
- 7. Data Validation Test – Change 2
 - 7.1 Create Table and Insert Data on New Primary
 - 7.2 Verify Data on Standby
- 8. Rebuild Old Primary (rac1) as Standby (Optional / Later Step)
 - 8.1 Stop PostgreSQL if Running
 - 8.2 Delete DATA, WAL, and ARCHIVE Files
 - 8.3 Perform Backup from Primary to Rebuild rac1 as Standby
 - 8.4 Verify postgresql.auto.conf and standby.signal
 - 8.5 Start PostgreSQL on rac1
- 9. Replication Verification – Change 3
 - 9.1 Load Test Data into Table on Primary (rac2)
 - 9.2 Monitor Replication on Standby 1 (rac1)

- 9.3 Monitor Replication on Standby 2 (rac3)
 - 10. Log Verification
 - 10.1 Check Logs on Primary
 - 10.2 Check Logs on Standby 1 (rac1)
 - 10.3 Check Logs on Standby 2 (rac3)
-

o. Objective : Perform Failover from rac1 → rac2

PostgreSQL natively supports **failover**, not **switchover** (unlike Oracle Data Guard).

For managed or automated switchover, use tools such as **repmgr** or **Patroni**.

Notes:

- In case of an **unplanned outage** or **disaster recovery**, if the primary becomes unreachable, **promote the standby** to act as the new primary.
- The old primary must then be **rebuilt as a standby** using data from the new primary.
- For production environments, **use replication slots** to prevent WAL loss.
- **Synchronous replication** → Zero data loss (commit waits for standby acknowledgment).
- **Asynchronous replication** → Possible data loss during failover.
- Always **check replication lag** before performing a planned failover.

1. Environment Setup

NODE	IP	ROLE	SYNC TYPE	PURPOSE
rac1	192.168.2.21	Primary	async	Old primary (will fail)
rac2	192.168.2.22	Standby	async	To be promoted as new primary
rac3	192.168.3.20	Standby	async	Will remain standby after failover

2. Simulate and Confirm Primary (rac1) Failure

DISCLAIMER: The kill command given below is just for learning purposes and should only be used on testing systems. We will not take any responsibility of any consequences or loss of data caused by this command.

Note: In this demonstration, I used a simple KILL command to simulate a PostgreSQL failover. In real-world scenarios, failover events may occur due to factors such as server unavailability, disk corruption, or other critical system failures.

```
[root@rac1 data]# ps -ef | grep postgres
postgres    12125          1  0 07:56 ?           00:00:00
/usr/pgsql-17/bin/postgres -D /pgData/pgsql17/data
postgres    12127      12125  0 07:56 ?           00:00:00 postgres:
logger
postgres    12128      12125  0 07:56 ?           00:00:00 postgres:
checkpointer
postgres    12129      12125  0 07:56 ?           00:00:00 postgres:
background writer
postgres    12131      12125  0 07:56 ?           00:00:00 postgres:
walwriter
postgres    12132      12125  0 07:56 ?           00:00:00 postgres:
autovacuum launcher
postgres    12133      12125  0 07:56 ?           00:00:00 postgres:
archiver last was 0000000100000000000000017.00000028.backup
postgres    12134      12125  0 07:56 ?           00:00:00 postgres:
logical replication launcher
postgres    12282      12125  0 08:08 ?           00:00:00 postgres:
walsender repuser 192.168.2.22(51152) streaming 0/18023510
postgres    12283      12125  0 08:08 ?           00:00:00 postgres:
walsender repuser 192.168.3.20(51354) streaming 0/18023510
root         15502       3678  0 12:52 pts/0      00:00:00 grep --
color=auto postgres
[root@rac1 data]# kill -9 12125
[root@rac1 data]# ps -ef | grep postgres
root         15513       3678  0 12:52 pts/0      00:00:00 grep --
color=auto postgres
[root@rac1 data]# cd
[root@rac1 ~]#

[postgres@rac1 ~]$ psql
psql: error: connection to server on socket
"/run/postgresql/.s.PGSQL.5432" failed: Connection refused
        Is the server running locally and accepting
connections on that socket?
[postgres@rac1 ~]$
```

3. Failover Procedure: Promote rac2 (Standby1) as New Primary

3.1 Promote rac2 to Primary

```
# use pg_ctl promote -D /pgData/pgsql17/data (or) psql -U postgres -c
"SELECT pg_promote();" to initiate failover

[postgres@rac2 ~]$ cat /pgData/pgsql17/data/log/postgresql-Fri.log
..
..
        Is the server running on that host and accepting TCP/IP
connections?
2025-10-17 12:57:50.553 EDT [9070] LOG:  waiting for WAL to become
available at 0/18023528
2025-10-17 12:57:55.552 EDT [11746] FATAL:  could not connect to the
primary server: connection to server at "192.168.2.21", port 5432 failed:
Connection refused
        Is the server running on that host and accepting TCP/IP
connections?
2025-10-17 12:57:55.554 EDT [9070] LOG:  waiting for WAL to become
available at 0/18023528
2025-10-17 12:58:00.559 EDT [11747] FATAL:  could not connect to the
primary server: connection to server at "192.168.2.21", port 5432 failed:
Connection refused
        Is the server running on that host and accepting TCP/IP
connections?
2025-10-17 12:58:00.562 EDT [9070] LOG:  waiting for WAL to become
available at 0/18023528
2025-10-17 12:58:05.562 EDT [11748] FATAL:  could not connect to the
primary server: connection to server at "192.168.2.21", port 5432 failed:
Connection refused
        Is the server running on that host and accepting TCP/IP
connections?
2025-10-17 12:58:05.564 EDT [9070] LOG:  waiting for WAL to become
available at 0/18023528
[postgres@rac2 ~]$

[postgres@rac2 ~]$ psql -U postgres -c "SELECT pg_promote();"
 pg_promote
-----
 t  <---- Failover completed successfully.
(1 row)
```

```
[postgres@rac2 ~]$
```

Verify logs

```
[postgres@rac2 ~]$ cat /pgData/pgsql17/data/log/postgresql-Fri.log
```

```
..  
..
```

```
2025-10-17 12:59:30.609 EDT [9070] LOG:  waiting for WAL to become  
available at 0/18023528
```

```
2025-10-17 12:59:32.986 EDT [9070] LOG:  received promote request
```

```
2025-10-17 12:59:32.986 EDT [9070] LOG:  redo done at 0/180234D8 system  
usage: CPU: user: 0.02 s, system: 0.02 s, elapsed: 17448.30 s
```

```
2025-10-17 12:59:32.986 EDT [9070] LOG:  last completed transaction was at  
log time 2025-10-17 08:32:12.184495-04
```

```
2025-10-17 12:59:32.987 EDT [9070] LOG:  selected new timeline ID: 2
```

```
2025-10-17 12:59:33.220 EDT [9070] LOG:  archive recovery complete
```

```
2025-10-17 12:59:33.257 EDT [9068] LOG:  checkpoint starting: force
```

```
2025-10-17 12:59:33.261 EDT [9064] LOG:  database system is ready to accept  
connections
```

```
2025-10-17 12:59:33.320 EDT [9068] LOG:  checkpoint complete: wrote 2  
buffers (0.0%); 0 WAL file(s) added, 0 removed, 0 recycled; write=0.001 s,  
sync=0.017 s, total=0.064 s; sync files=2, longest=0.012 s, average=0.009  
s; distance=0 kB, estimate=39825 kB; lsn=0/180235A0, redo lsn=0/18023548
```

```
2025-10-17 13:04:33.420 EDT [9068] LOG:  checkpoint starting: time
```

```
2025-10-17 13:04:33.543 EDT [9068] LOG:  checkpoint complete: wrote 1  
buffers (0.0%); 0 WAL file(s) added, 0 removed, 0 recycled; write=0.101 s,  
sync=0.005 s, total=0.123 s; sync files=1, longest=0.005 s, average=0.005  
s; distance=1 kB, estimate=35843 kB; lsn=0/18023C68, redo lsn=0/18023C10
```

```
[postgres@rac2 ~]$
```

3.2 Update pg_hba.conf

```
# Update pg_hba.conf with standby servers IP's
```

```
FROM:
```

```
[postgres@rac2 ~]$ cat /pgData/pgsql17/data/pg_hba.conf
```

```
..  
..
```

```
# Allow standby servers for replication
```

```
host      replication      repuser      192.168.2.22/32      scram-sha-256
```

```
host      replication      repuser      192.168.3.20/32      scram-sha-256
```

```
[postgres@rac2 ~]$
```

```
TO:
```

```
[postgres@rac2 ~]$ cat /pgData/pgsql17/data/pg_hba.conf
..
..
# Allow standby servers for replication
host      replication    repuser      192.168.2.21/32      scram-sha-256
host      replication    repuser      192.168.3.20/32      scram-sha-256
[postgres@rac2 ~]$
```

3.3 Reset primary_conninfo in postgresql.auto.conf

```
# Update primary_conninfo info on postgresql.auto.conf
# Primary node will not have primary_conninfo entires, hence we are
removing from primary node.

FROM:

[postgres@rac2 ~]$ cat /pgData/pgsql17/data/postgresql.auto.conf
# Do not edit this file manually!
# It will be overwritten by the ALTER SYSTEM command.
primary_conninfo = 'user=repuser password=repuser channel_binding=prefer
host=192.168.2.21 port=5432 sslmode=prefer sslnegotiation=postgres
sslcompression=0 sslcertmode=allow sslni=1
ssl_min_protocol_version=TLsv1.2 gssencmode=prefer krbsrvname=postgres
gssdelegation=0 target_session_attrs=any load_balance_hosts=disable'
primary_slot_name = 'rac2_standby_slot'
[postgres@rac2 ~]$
```

```
TO:

[postgres@rac2 ~]$ psql -U postgres -c "ALTER SYSTEM RESET
primary_conninfo;"
ALTER SYSTEM
[postgres@rac2 ~]$ psql -U postgres -c "ALTER SYSTEM RESET
primary_slot_name;"
ALTER SYSTEM
[postgres@rac2 ~]$ psql -U postgres -c "SELECT pg_reload_conf();"
pg_reload_conf
-----
t
(1 row)
```

```
[postgres@rac2 ~]$  
  
[postgres@rac2 ~]$ cat /pgData/pgsql17/data/postgresql.auto.conf  
# Do not edit this file manually!  
# It will be overwritten by the ALTER SYSTEM command.  
[postgres@rac2 ~]$
```

3.4 Restart PostgreSQL Service

```
[postgres@rac2 ~]$ psql -U postgres -c "SELECT pg_reload_conf();"
pg_reload_conf
-----
t
(1 row)

[postgres@rac2 ~]$  
  
--- OR ---  
  
[root@rac2 ~]# systemctl stop postgresql-17.service  
[root@rac2 ~]# systemctl start postgresql-17.service
```

4. Verify Failover Operation (rac2 is primary)

```
# Verify that rac2 is now primary  
  
[postgres@rac2 ~]$ psql -U postgres -c "SELECT pg_is_in_recovery();"
pg_is_in_recovery
-----
f <---- false means Primary  
(1 row)  
  
[postgres@rac2 ~]$
```

5. Reconfigure rac3 (Standby2) to Point to New Primary (rac2)

5.1 Update primary_conninfo on rac3

```

# Update primary_conninfo info on postgresql.auto.conf

FROM:
[postgres@rac3 ~]$ cat /pgData/pgsql17/data/postgresql.auto.conf
# Do not edit this file manually!
# It will be overwritten by the ALTER SYSTEM command.
primary_conninfo = 'user=repuser password=repuser channel_binding=prefer
host=192.168.2.21 port=5432 sslmode=prefer sslnegotiation=postgres
sslcompression=0 sslcertmode=allow sslsni=1
ssl_min_protocol_version=TLSv1.2 gssencmode=prefer krbsrvname=postgres
gssdelegation=0 target_session_attrs=any load_balance_hosts=disable'
primary_slot_name = 'rac3_standby_slot'
[postgres@rac3 ~]$

TO:

postgres=# ALTER SYSTEM SET primary_conninfo = 'user=repuser
password=repuser channel_binding=prefer host=192.168.2.22 port=5432
sslmode=prefer sslnegotiation=postgres sslcompression=0 sslcertmode=allow
sslsni=1 ssl_min_protocol_version=TLSv1.2 gssencmode=prefer
krbsrvname=postgres gssdelegation=0 target_session_attrs=any
load_balance_hosts=disable';
ALTER SYSTEM
postgres=# ALTER SYSTEM SET primary_slot_name = 'rac3_standby_slot';
ALTER SYSTEM
postgres=#
postgres=# SELECT pg_reload_conf();
pg_reload_conf
-----
t
(1 row)

postgres=#

[postgres@rac3 ~]$ cat /pgData/pgsql17/data/postgresql.auto.conf
# Do not edit this file manually!
# It will be overwritten by the ALTER SYSTEM command.
primary_conninfo = 'user=repuser password=repuser channel_binding=prefer
host=192.168.2.22 port=5432 sslmode=prefer sslnegotiation=postgres
sslcompression=0 sslcertmode=allow sslsni=1
ssl_min_protocol_version=TLSv1.2 gssencmode=prefer krbsrvname=postgres
gssdelegation=0 target_session_attrs=any load_balance_hosts=disable'
primary_slot_name = 'rac3_standby_slot'
[postgres@rac3 ~]$

[postgres@rac3 ~]$ psql -U postgres -c "SELECT pg_is_in_recovery();"

```



```
pg_is_in_recovery
-----
t  <---- True means Standby
(1 row)

[postgres@rac3 ~]$
```

5.2 Update pg_hba.conf

```
[postgres@rac3 ~]$ cat /pgData/pgsql17/data/log/postgresql-Fri.log
..
..
        Is the server running on that host and accepting TCP/IP
connections?
2025-10-17 13:00:30.625 EDT [8688] LOG:  waiting for WAL to become
available at 0/18023528
2025-10-17 13:00:35.627 EDT [11404] FATAL:  could not connect to the
primary server: connection to server at "192.168.2.21", port 5432 failed:
Connection refused
        Is the server running on that host and accepting TCP/IP
connections?
2025-10-17 13:00:35.629 EDT [8688] LOG:  waiting for WAL to become
available at 0/18023528
[postgres@rac3 ~]$

# Update pg_hba.conf

FROM:

[postgres@rac3 ~]$ cat /pgData/pgsql17/data/pg_hba.conf
..
..
# Allow standby servers for replication
host      replication    repuser    192.168.2.22/32    scram-sha-256
host      replication    repuser    192.168.3.20/32    scram-sha-256
[postgres@rac3 ~]$

TO:

[postgres@rac3 ~]$ cat /pgData/pgsql17/data/pg_hba.conf
..
..
# Allow standby servers for replication
host      replication    repuser    192.168.2.21/32    scram-sha-256
```

```
host      replication    repuser    192.168.3.20/32    scram-sha-256
[postgres@rac3 ~]$
```

5.3 Restart PostgreSQL Service

```
[postgres@rac3 ~]$ psql -U postgres -c "SELECT pg_reload_conf();"
pg_reload_conf
-----
t
(1 row)

[postgres@rac3 ~]$

--- OR ---

[root@rac3 ~]# systemctl stop postgresql-17.service
[root@rac3 ~]# systemctl start postgresql-17.service
```

6. Data Validation Test – Change 1

6.1 Insert Data on New Primary and Verify logs

On Primary (rac2)

```
[postgres@rac2 ~]$ psql
psql (17.6)
Type "help" for help.

postgres=# SELECT * FROM EMP;
 name | designation | project | company 
-----+-----+-----+-----
 Sugi | DBA         | Jetstar | iGATE
 Teja | DBA         | RCM     | iGATE
 RAJ  | DBA         | RCM     | iGATE
(3 rows)

postgres=# INSERT INTO emp VALUES ('KK', 'DBA LEAD', 'RCM', 'iGATE');
INSERT 0 1
postgres=# SELECT * FROM EMP;
 name | designation | project | company 
-----+-----+-----+-----
 Sugi | DBA         | Jetstar | iGATE
```

```

Teja | DBA          | RCM      | iGATE
RAJ  | DBA          | RCM      | iGATE
KK   | DBA LEAD     | RCM      | iGATE <-----
(4 rows)

```

```
postgres=#
```

```

2025-10-17 13:57:30.510 EDT [12525] ERROR:  replication slot
"rac3_standby_slot" does not exist
2025-10-17 13:57:30.510 EDT [12525] STATEMENT:  START_REPLICATION SLOT
"rac3_standby_slot" 0/18000000 TIMELINE 2

```

6.2 Create replication slot (rac3_standby_slot) on New Primary

```
# On new primary
```

```
[postgres@rac2 ~]$ psql -U postgres -c "SELECT * FROM
pg_create_physical_replication_slot('rac3_standby_slot');"
```

```

      slot_name      | lsn
-----+-----
rac3_standby_slot |
(1 row)

```

```
[postgres@rac2 ~]$
```

```

2025-10-17 13:59:30.571 EDT [12570] ERROR:  replication slot
"rac3_standby_slot" does not exist
2025-10-17 13:59:30.571 EDT [12570] STATEMENT:  START_REPLICATION SLOT
"rac3_standby_slot" 0/18000000 TIMELINE 2
2025-10-17 13:59:34.554 EDT [9068] LOG:  checkpoint starting: time
2025-10-17 13:59:34.681 EDT [9068] LOG:  checkpoint complete: wrote 2
buffers (0.0%); 0 WAL file(s) added, 0 removed, 0 recycled; write=0.102 s,
sync=0.006 s, total=0.127 s; sync files=2, longest=0.004 s, average=0.003
s; distance=0 kB, estimate=32259 kB; lsn=0/18023EF0, redo lsn=0/18023E98
2025-10-17 13:59:35.572 EDT [12571] ERROR:  replication slot
"rac3_standby_slot" does not exist
2025-10-17 13:59:35.572 EDT [12571] STATEMENT:  START_REPLICATION SLOT
"rac3_standby_slot" 0/18000000 TIMELINE 2
2025-10-17 13:59:40.577 EDT [12572] ERROR:  replication slot
"rac3_standby_slot" does not exist
2025-10-17 13:59:40.577 EDT [12572] STATEMENT:  START_REPLICATION SLOT
"rac3_standby_slot" 0/18000000 TIMELINE 2
2025-10-17 13:59:45.580 EDT [12574] ERROR:  replication slot

```

```
"rac3_standby_slot" does not exist
2025-10-17 13:59:45.580 EDT [12574] STATEMENT:  START_REPLICATION SLOT
"rac3_standby_slot" 0/18000000 TIMELINE 2
2025-10-17 13:59:50.581 EDT [12575] ERROR:  replication slot
"rac3_standby_slot" does not exist
2025-10-17 13:59:50.581 EDT [12575] STATEMENT:  START_REPLICATION SLOT
"rac3_standby_slot" 0/18000000 TIMELINE 2
[postgres@rac2 ~]$

[postgres@rac2 ~]$ psql -U postgres -c "
> SELECT
> application_name AS standby_name,
> client_addr AS standby_ip,
> state,
> sync_state,
> pg_wal_lsn_diff(pg_current_wal_lsn(), replay_lsn) AS byte_lag
> FROM pg_stat_replication;
> "
 standby_name | standby_ip | state   | sync_state | byte_lag
-----+-----+-----+-----+-----
 walreceiver  | 192.168.3.20 | streaming | async      |          0 <--- no
lag
(1 row)

[postgres@rac2 ~]$

*** Aftrer few seconds there were no errors reported in the logs.
```

6.3 Verify Data on Standby and logs

```
postgres=# SELECT * FROM EMP;
 name | designation | project | company
-----+-----+-----+-----
 Sugi | DBA         | Jetstar | iGATE
 Teja | DBA         | RCM     | iGATE
 RAJ  | DBA         | RCM     | iGATE
 KK   | DBA LEAD   | RCM     | iGATE <----
(4 rows)

postgres=#

2025-10-17 13:59:50.581 EDT [12669] FATAL:  could not start WAL streaming:
ERROR:  replication slot "rac3_standby_slot" does not exist
2025-10-17 13:59:50.584 EDT [8688] LOG:  waiting for WAL to become
```

```

available at 0/18002000
2025-10-17 13:59:55.583 EDT [12670] LOG:  started streaming WAL from
primary at 0/18000000 on timeline 2
2025-10-17 13:59:57.951 EDT [8686] LOG:  restartpoint starting: time
2025-10-17 13:59:58.183 EDT [8686] LOG:  restartpoint complete: wrote 3
buffers (0.0%); 0 WAL file(s) added, 0 removed, 0 recycled; write=0.207 s,
sync=0.009 s, total=0.232 s; sync files=3, longest=0.005 s, average=0.003
s; distance=2 kB, estimate=13284 kB; lsn=0/18023EF0, redo lsn=0/18023E98
2025-10-17 13:59:58.183 EDT [8686] LOG:  recovery restart point at
0/18023E98
2025-10-17 13:59:58.183 EDT [8686] DETAIL:  Last completed transaction was
at log time 2025-10-17 13:55:47.944361-04.
[postgres@rac3 ~]$

*** After few seconds there were no errors reported in the logs.

```

7. Data Validation Test – Change 2

7.1 Create Table and Insert Data on New Primary

```

[postgres@rac2 ~]$ psql
psql (17.6)
Type "help" for help.

postgres=# CREATE TABLE test_failover(id serial primary key, msg text);
CREATE TABLE
postgres=# INSERT INTO test_failover(msg) VALUES ('Replication working
after failover');
INSERT 0 1
postgres=#
postgres=#
postgres=# SELECT * FROM test_failover;
 id |          msg
-----+-----
  1 | Replication working after failover
(1 row)

postgres=#

[postgres@rac2 ~]$ cat /pgData/pgsql17/data/log/postgresql-Fri.log
..
..
2025-10-17 14:29:35.647 EDT [9068] LOG:  checkpoint starting: time
2025-10-17 14:29:40.208 EDT [9068] LOG:  checkpoint complete: wrote 45

```

```
buffers (0.3%); 0 WAL file(s) added, 0 removed, 0 recycled; write=4.436 s,
sync=0.092 s, total=4.561 s; sync files=38, longest=0.037 s, average=0.003
s; distance=184 kB, estimate=26162 kB; lsn=0/18077EA8, redo lsn=0/18077E50
[postgres@rac2 ~]$
```

*** No errors reported in the logs.

7.2 Verify Data on Standby

```
[postgres@rac3 ~]$ psql -U postgres -c "SELECT * FROM test_failover;"
```

```
id |          msg
----+-----
  1 | Replication working after failover <---- Working as expected.
(1 row)
```

```
[postgres@rac3 ~]$
```

```
[postgres@rac3 ~]$ cat /pgData/pgsql17/data/log/postgresql-Fri.log
```

```
..
..
```

```
2025-10-17 14:29:52.637 EDT [12847] LOG:  restartpoint starting: time
2025-10-17 14:29:57.261 EDT [12847] LOG:  restartpoint complete: wrote 46
buffers (0.3%); 0 WAL file(s) added, 0 removed, 0 recycled; write=4.537 s,
sync=0.072 s, total=4.624 s; sync files=38, longest=0.015 s, average=0.002
s; distance=184 kB, estimate=184 kB; lsn=0/18077EA8, redo lsn=0/18077E50
```

```
2025-10-17 14:29:57.261 EDT [12847] LOG:  recovery restart point at
0/18077E50
```

```
2025-10-17 14:29:57.261 EDT [12847] DETAIL:  Last completed transaction was
at log time 2025-10-17 14:28:28.964977-04.
```

```
[postgres@rac3 ~]$
```

*** No errors reported in the logs.

8. Rebuild Old Primary (rac1) as Standby (Optional / Later Step)

Rebuild using pg_basebackup (Clean and reliable, always works)

8.1 Stop PostgreSQL if Running

```
[root@rac1 ~]# systemctl stop postgresql-17.service
[root@rac1 ~]#
[root@rac1 ~]# ps -ef | grep postgres
root      16544    3678  0 14:43 pts/0    00:00:00 grep --color=auto
postgres
[root@rac1 ~]#
```

8.2 Delete DATA, WAL, and ARCHIVE Files

```
[root@rac1 ~]# su - postgres
[postgres@rac1 ~]$
[postgres@rac1 ~]$ rm -rf /pgData/pgsql17/data/*
[postgres@rac1 ~]$ rm -rf /pgWal/pgsql17/wal/*
[postgres@rac1 ~]$ rm -rf /pgArch/pgsql17/arch/*
[postgres@rac1 ~]$
```

8.3 Perform Backup from Primary to Rebuild rac1 as Standby

```
[postgres@rac1 ~]$ pg_basebackup -h 192.168.2.22 -U repuser -p 5432 -D
/pgData/pgsql17/data -Fp -Xs -P -R -C -S rac1_standby_slot
Password:
24643/24643 kB (100%), 1/1 tablespace
[postgres@rac1 ~]$
```

8.4 Verify postgresql.auto.conf and standby.signal

```
[postgres@rac1 ~]$ ls -ltr /pgData/pgsql17/data/standby.signal
-rw-----. 1 postgres postgres 0 Oct 17 14:49
/pgData/pgsql17/data/standby.signal
[postgres@rac1 ~]$
[postgres@rac1 ~]$ cat /pgData/pgsql17/data/postgresql.auto.conf
# Do not edit this file manually!
# It will be overwritten by the ALTER SYSTEM command.
primary_conninfo = 'user=repuser password=repuser channel_binding=prefer
host=192.168.2.22 port=5432 sslmode=prefer sslnegotiation=postgres
sslcompression=0 sslcertmode=allow sslsni=1
ssl_min_protocol_version=TLSv1.2 gssencmode=prefer krbsrvname=postgres
gssdelegation=0 target_session_attrs=any load_balance_hosts=disable'
```

```
primary_slot_name = 'rac1_standby_slot'
[postgres@rac1 ~]$
```

8.5 Start PostgreSQL on rac1

```
[root@rac1 ~]# systemctl start postgresql-17.service
[root@rac1 ~]#

[postgres@rac1 ~]$ cat /pgData/pgsql17/data/log/postgresql-Fri.log
..
..
2025-10-17 14:54:36.566 EDT [16742] LOG:  starting PostgreSQL 17.6 on
x86_64-pc-linux-gnu, compiled by gcc (GCC) 8.5.0 20210514 (Red Hat 8.5.0-
26), 64-bit
2025-10-17 14:54:36.566 EDT [16742] LOG:  listening on IPv4 address
"0.0.0.0", port 5432
2025-10-17 14:54:36.573 EDT [16742] LOG:  listening on Unix socket
"/run/postgresql/.s.PGSQL.5432"
2025-10-17 14:54:36.586 EDT [16742] LOG:  listening on Unix socket
"/tmp/.s.PGSQL.5432"
2025-10-17 14:54:36.596 EDT [16748] LOG:  database system was interrupted;
last known up at 2025-10-17 14:49:32 EDT
2025-10-17 14:54:38.095 EDT [16748] LOG:  starting backup recovery with
redo LSN 0/19000028, checkpoint LSN 0/19000080, on timeline ID 2
2025-10-17 14:54:38.095 EDT [16748] LOG:  entering standby mode
2025-10-17 14:54:38.106 EDT [16748] LOG:  redo starts at 0/19000028
2025-10-17 14:54:38.111 EDT [16748] LOG:  completed backup recovery with
redo LSN 0/19000028 and end LSN 0/19000120
2025-10-17 14:54:38.111 EDT [16748] LOG:  consistent recovery state reached
at 0/19000120
2025-10-17 14:54:38.112 EDT [16742] LOG:  database system is ready to
accept read-only connections
2025-10-17 14:54:38.181 EDT [16749] LOG:  started streaming WAL from
primary at 0/1A000000 on timeline 2
[postgres@rac1 ~]$

*** No errors reported in the logs.
```

9. Replication Verification – Change 3

9.1 Load Test Data into Table on Primary (rac2)


```
# Verify lag from Primary node
```

```
[postgres@rac2 ~]$ psql -U postgres -c "
```

```
> SELECT
```

```
> application_name AS standby_name,
```

```
> client_addr AS standby_ip,
```

```
> state,
```

```
> sync_state,
```

```
> pg_wal_lsn_diff(pg_current_wal_lsn(), replay_lsn) AS byte_lag
```

```
> FROM pg_stat_replication;
```

```
> "
```

standby_name	standby_ip	state	sync_state	byte_lag
walreceiver	192.168.2.21	streaming	async	0
walreceiver	192.168.3.20	streaming	async	0

(2 rows)

```
[postgres@rac2 ~]$
```

```
[postgres@rac2 ~]$ psql
```

```
psql (17.6)
```

```
Type "help" for help.
```

```
postgres=# \x
```

```
Expanded display is on.
```

```
postgres=# select * from pg_stat_replication;
```

```
-[ RECORD 1 ]-----+-----
```

pid	13145
usesysid	16452
username	repuser
application_name	walreceiver
client_addr	192.168.2.21
client_hostname	
client_port	40334
backend_start	2025-10-17 14:54:38.15459-04
backend_xmin	
state	streaming
sent_lsn	0/1A000430
write_lsn	0/1A000430
flush_lsn	0/1A000430
replay_lsn	0/1A000430
write_lag	
flush_lag	
replay_lag	
sync_priority	0
sync_state	async
reply_time	2025-10-17 17:05:39.286255-04

```

-[ RECORD 2 ]-----+-----
pid           | 12747
usesysid      | 16452
username      | repuser
application_name | walreceiver
client_addr   | 192.168.3.20
client_hostname |
client_port   | 36018
backend_start | 2025-10-17 14:16:07.947815-04
backend_xmin  |
state         | streaming
sent_lsn      | 0/1A000430
write_lsn     | 0/1A000430
flush_lsn     | 0/1A000430
replay_lsn    | 0/1A000430
write_lag     |
flush_lag     |
replay_lag    |
sync_priority | 0
sync_state    | async
reply_time    | 2025-10-17 17:05:39.320194-04

```

postgres=#

Verify Replication slots

```
[postgres@rac2 ~]$ psql -U postgres -c "SELECT slot_name, slot_type,
database, active FROM pg_replication_slots;"
```

slot_name	slot_type	database	active
rac3_standby_slot	physical		t
rac1_standby_slot	physical		t

(2 rows)

[postgres@rac2 ~]\$

```
[postgres@rac2 ~]$ psql -U postgres -c "SELECT * FROM EMP;"
```

name	designation	project	company
Sugi	DBA	Jetstar	iGATE
Teja	DBA	RCM	iGATE
RAJ	DBA	RCM	iGATE
KK	DBA LEAD	RCM	iGATE

(4 rows)

```
[postgres@rac2 ~]$ psql -U postgres -c "INSERT INTO emp VALUES ('DEV', 'DBA
MANAGER', 'RCM', 'iGATE');"
```

```

INSERT 0 1
[postgres@rac2 ~]$
[postgres@rac2 ~]$ psql -U postgres -c "SELECT * FROM
EMP;"
  name | designation | project | company
-----+-----+-----+-----
  Sugi | DBA         | Jetstar | iGATE
  Teja | DBA         | RCM     | iGATE
  RAJ  | DBA         | RCM     | iGATE
  KK   | DBA LEAD   | RCM     | iGATE
  DEV  | DBA MANAGER | RCM     | iGATE <---
(5 rows)

[postgres@rac2 ~]$

```

9.2 Monitor Replication on Standby 1 (rac1)

```

[postgres@rac1 ~]$ psql -U postgres -c "SELECT slot_name, slot_type,
database, active FROM pg_replication_slots;"
 slot_name | slot_type | database | active
-----+-----+-----+-----
(0 rows)

[postgres@rac1 ~]$

[postgres@rac1 ~]$ psql -U postgres -c "SELECT * FROM
EMP;"
  name | designation | project | company
-----+-----+-----+-----
  Sugi | DBA         | Jetstar | iGATE
  Teja | DBA         | RCM     | iGATE
  RAJ  | DBA         | RCM     | iGATE
  KK   | DBA LEAD   | RCM     | iGATE
  DEV  | DBA MANAGER | RCM     | iGATE <---
(5 rows)

[postgres@rac1 ~]$

```

9.3 Monitor Replication on Standby 2 (rac3)

```
[postgres@rac3 ~]$ psql -U postgres -c "SELECT slot_name, slot_type,
database, active FROM pg_replication_slots;"
 slot_name | slot_type | database | active
-----+-----+-----+-----
(0 rows)

[postgres@rac3 ~]$
[postgres@rac3 ~]$ psql -U postgres -c "SELECT * FROM EMP;"
 name | designation | project | company
-----+-----+-----+-----
 Sugi | DBA         | Jetstar | iGATE
 Teja | DBA         | RCM     | iGATE
 RAJ  | DBA         | RCM     | iGATE
 KK   | DBA LEAD    | RCM     | iGATE
 DEV  | DBA MANAGER | RCM     | iGATE <---
(5 rows)

[postgres@rac3 ~]$
```

10. Log Verification

10.1 Check Logs on Primary

```
[postgres@rac2 ~]$ tail -100 /pgData/pgsql17/data/log/postgresql-Fri.log
..
..
2025-10-17 14:54:32.813 EDT [9068] LOG:  checkpoint starting: time
2025-10-17 14:54:32.860 EDT [9068] LOG:  checkpoint complete: wrote 0
buffers (0.0%); 0 WAL file(s) added, 0 removed, 1 recycled; write=0.001 s,
sync=0.001 s, total=0.048 s; sync files=0, longest=0.000 s, average=0.000
s; distance=16384 kB, estimate=24261 kB; lsn=0/1A0000B8, redo
lsn=0/1A000060
2025-10-17 15:04:33.059 EDT [9068] LOG:  checkpoint starting: time
2025-10-17 15:04:33.219 EDT [9068] LOG:  checkpoint complete: wrote 2
buffers (0.0%); 0 WAL file(s) added, 0 removed, 2 recycled; write=0.101 s,
sync=0.007 s, total=0.160 s; sync files=2, longest=0.005 s, average=0.004
s; distance=0 kB, estimate=21835 kB; lsn=0/1A000380, redo lsn=0/1A000328
[postgres@rac2 ~]$
```

```
*** No errors reported in the logs.
```

10.2 Check Logs on Standby 1 (rac1)

```
[postgres@rac1 ~]$ tail -100 /pgData/pgsql17/data/log/postgresql-Fri.log
..
..
2025-10-17 15:04:36.818 EDT [16746] LOG:  restartpoint starting: time
2025-10-17 15:04:36.946 EDT [16746] LOG:  restartpoint complete: wrote 2
buffers (0.0%); 0 WAL file(s) added, 0 removed, 0 recycled; write=0.107 s,
sync=0.007 s, total=0.129 s; sync files=2, longest=0.005 s, average=0.004
s; distance=0 kB, estimate=14745 kB; lsn=0/1A000380, redo lsn=0/1A000328
2025-10-17 15:04:36.946 EDT [16746] LOG:  recovery restart point at
0/1A000328
2025-10-17 15:04:36.946 EDT [16746] DETAIL:  Last completed transaction was
at log time 2025-10-17 15:03:38.479334-04.
[postgres@rac1 ~]$

*** No errors reported in the logs.
```

10.3 Check Logs on Standby 2 (rac3)

```
[postgres@rac3 ~]$ tail -100 /pgData/pgsql17/data/log/postgresql-Fri.log
..
..
..
2025-10-17 15:04:38.825 EDT [12847] LOG:  restartpoint starting: time
2025-10-17 15:04:38.950 EDT [12847] LOG:  restartpoint complete: wrote 2
buffers (0.0%); 0 WAL file(s) added, 0 removed, 0 recycled; write=0.102 s,
sync=0.008 s, total=0.125 s; sync files=2, longest=0.005 s, average=0.004
s; distance=0 kB, estimate=14745 kB; lsn=0/1A000380, redo lsn=0/1A000328
2025-10-17 15:04:38.950 EDT [12847] LOG:  recovery restart point at
0/1A000328
2025-10-17 15:04:38.950 EDT [12847] DETAIL:  Last completed transaction was
at log time 2025-10-17 15:03:38.479334-04.
[postgres@rac3 ~]$

*** No errors reported in the logs.
```

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guarantee that it will work for you. Ensure that you run it in your test environment before using.

Thank you,

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